VOILA: Query-Driven Representation Selection for VLM Analytics over Multi-Tier Image Data

Rahul Atul Bhope*, K. R. Jayaram†, Vinod Muthusamy†,
Ritesh Kumar†, Vatche Isahagian†, Nalini Venkatasubramanian*
*University of California, Irvine †IBM Research

***Abstract*—Data systems increasingly answer natural-language questions over image collections by invoking vision–language models (VLMs). An image is rarely stored once. Ingestion pipelines keep a caption, thumbnails, compressed previews, and the original, spread across device, edge, cloud, and archive tiers. Yet, typically, a query reads the original from wherever it lives. It pays the full transfer, retrieval, and inference cost before anyone knows whether that detail was needed. We argue that choosing *which representation to read* is a query-optimization decision. A data system should make it before any pixel is accessed. We present VOILA, a representation-aware query layer. VOILA maintains a representation catalog and learns per-model success estimators from the query text. It calibrates the estimates so that they are comparable across representations. It then selects representations per query or per workload, under byte, latency, and tier-dependent retrieval-cost objectives and workload budgets. We evaluate VOILA on 1,445 disaster-response questions over aerial imagery with six VLMs (7B–235B parameters). On average, VOILA is as accurate as the best fixed representation chosen on validation data (70.5% vs. 70.4%; per model, from 1.7 points below to 3.8 points above) while reading 84% fewer bytes. Compared with always reading originals, it is 7.3 points more accurate and reads 97.5% fewer bytes. Under recorded cellular uplinks, the share of questions answered correctly within one second rises from 37.9% to 70.3%. On VQA-v2, GQA, and TextVQA, VOILA reduces tier-weighted access cost by 65–73% relative to reading originals, at 1.8–10.6 points lower accuracy. Query-driven selection also has a clear failure mode. Unseen question templates cost up to 49 accuracy points. On templated workloads, routing unprofiled templates to a validated fallback representation recovers the best fixed representation's accuracy.**

***Index Terms*—multimodal data management, query optimization, vision–language models, tiered storage, physical design**

I. INTRODUCTION

Vision–language models (VLMs) have made it possible to query image collections using natural language. An analyst might ask how many buildings are flooded in an aerial image, an insurer might assess roof damage from a claim photo, or an AI assistant might recall what a sign in a user's photo said. Such queries are increasingly supported by semantic query engines [1]–[4], multimodal retrieval-augmented generation [5]–[7], and agentic memory systems. Despite their different purposes, these systems generally treat an image as a single attribute that is retrieved and passed to a model for interpretation.

This abstraction overlooks how images are stored in practice. Ingestion pipelines often produce multiple representations of the same image, including captions, thumbnails, and compressed previews. These representations can differ in size by four orders of magnitude and may reside on different storage tiers, ranging from device or edge caches to cold object storage and archives [8]–[11]. Different questions, however, require different levels of visual detail. A caption or thumbnail may suffice for some queries, while others require the original image. Yet existing systems often retrieve the original regardless of what the question requires.

TABLE I
FIXED-REPRESENTATION ACCURACY (%) ON THE 1,445-QUESTION FLOODNET POPULATION AND MEASURED REPRESENTATION SIZES. BOLD MARKS THE BEST FIXED INPUT PER MODEL. THE ORIGINAL IS NEVER THE MOST ACCURATE INPUT; THE GAP IS SIGNIFICANT FOR EVERY MODEL EXCEPT LLAVA-1.5-7B.

	Caption	32px	JPEG Q1	JPEG Q10	Original
Mean size (bytes)	281	2,141	217,576	445,479	5,520,774
Llama-4-Mav.	54.2	24.6	64.7	**70.2**	63.1
LLaVA-1.5-7B	73.3	73.8	42.9	**75.6**	74.7
Pixtral-12B	17.7	20.2	28.0	**50.5**	38.8
Qwen3-VL-235B	66.0	25.8	75.9	**83.2**	76.6
Qwen2-VL-72B	30.7	**73.9**	23.6	39.8	58.8
Qwen2.5-VL-7B	42.8	59.7	64.5	**71.5**	67.4

This default can be expensive, particularly when the original image must be fetched from a remote or archival storage tier. More surprisingly, it is not always the most accurate choice. For some questions, a smaller representation can produce a better answer than the original image. Section II demonstrates both effects using a disaster-response workload.

An alternative is to select a representation based on the question being asked. Existing systems largely leave this decision to the application (§II-D), but making it automatically poses two challenges. First, a natural-language question does not explicitly specify the visual detail needed to answer it. Second, the decision must be made before retrieving the image. Inspecting the original to determine which representation is sufficient would defeat the purpose of avoiding that retrieval.

Our approach. We present VOILA, which makes representation selection a cost-based query optimization decision. Our key observation is that the wording and structure of a question provide a useful prior about the visual evidence needed to answer it (§II-C). We use this observation to estimate, from the question alone, the probability that a given VLM will answer correctly using each available representation. These estimates are calibrated separately for each model and representation,

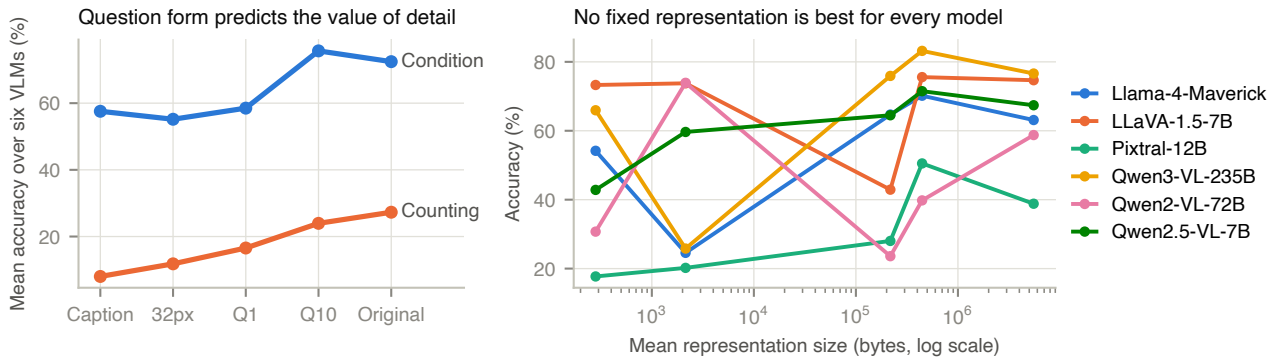


Fig. 1. Accuracy by representation on the FloodNet workload. Left: questions grouped by surface form and averaged over six VLMs. Counting questions benefit from increasing detail, while condition-recognition questions peak at a compressed JPEG preview. Right: accuracy is non-monotone in representation size, and models differ in their preferred input.

and combined with access costs obtained from a representation catalog. The resulting optimizer selects a representation without first retrieving the original image. The same estimates support three optimization problems: per-query selection (P1), workload allocation under a shared budget (P2), and representation placement across storage tiers (P3).

Results. We evaluate VOILA on 1,445 disaster-response questions using six VLMs. On average, it is as accurate as the best fixed representation chosen on validation data (70.5% vs. 70.4%) while reading 84% fewer bytes. Per model, the paired difference ranges from 1.7 points below to 3.8 points above, and is not significant for four of the six models. Under recorded cellular uplinks, it nearly doubles the fraction of questions answered correctly within one second. On three general VQA benchmarks, it reduces tier-weighted access cost by 65–73% relative to always reading the original, at 1.8–10.6 points lower accuracy (Table IV). Under shared byte budgets, workload allocation is on average 1.6–3.1 points more accurate than the best fixed representation that fits. In a two-tier simulation, placement guided by the same estimates answers 68.5% of questions within a 0.25 s read deadline with 0.5% of the corpus at the edge, against at most 48.5% for frequency-based tiering of originals. Its main weakness is question templates absent from profiling. They cost up to 49 accuracy points, even when worded like profiled ones. On templated workloads, falling back to the best fixed representation recovers that loss.

Contributions:

- We formulate *query-driven representation selection* for VLM queries over multi-representation, multi-tier image data. We study three problems: per-query selection (P1), workload allocation under a shared budget (P2), and representation placement across storage tiers (P3) (§III).
- We design VOILA, a representation-aware query layer that combines a representation catalog, calibrated per-model success estimators, and cost-based optimizers. We bound the regret of per-query selection in terms of estimator error and reduce workload allocation to a multiple-choice knapsack problem. For P3 we give a greedy placement advisor rather than an optimal solution (§IV).
- We evaluate VOILA across six VLMs and four datasets, using measured payload sizes, tier-weighted access costs, recorded network traces, shared workload budgets, and a two-tier placement simulation. Our evaluation uses image-grouped protocols and clustered confidence intervals. We identify cases where fixed representations and supervised routers remain competitive, as well as scenarios where query-driven selection fails (§VI).

II. MOTIVATING USE CASE AND THE GAP

A. Post-disaster damage assessment

Following a hurricane, survey teams use UAVs to capture high-resolution images of affected neighborhoods [12]. Images are captured faster than they can be transferred over degraded networks. They are stored locally on the capture device and copied to a field gateway, which connects to cloud storage over a cellular or satellite backhaul. Over time, the original images may move to colder storage tiers.

Analysts at an emergency operations center query these images to assess damage and plan response efforts. They might ask *"What is the overall condition of this area?"*, *"Is the road flooded?"*, or *"How many buildings are non-flooded?"* Automated agents may issue similar queries, with a cloud-hosted VLM generating the answers.

The FloodNet benchmark [12] captures this type of workload. Its questions fall into four classes: condition recognition, yes/no, simple counting, and complex counting. The workload presents competing requirements. Answers must be accurate enough to support response decisions, but they must also arrive quickly. The backhaul is bandwidth-constrained and shared with voice and coordination traffic, while cloud retrieval and VLM inference incur additional costs. The choice of which image representation to retrieve, and from which storage tier, therefore directly affects both response quality and cost.

B. Why fixed representations fall short

Table I shows the five representations available for each FloodNet image, their average sizes, and the accuracy achieved

when each VLM uses a fixed representation for every question. The representations range from a 281-byte caption to a 5.5 MB original image. The results expose the limitations of several straightforward approaches.

Always retrieve the original. This transfers approximately 8 GB for the 1,445-question workload, or 5.5 MB per question, and achieves a mean accuracy of 63.2% across the six models. Under recorded cellular uplinks, only 37.9% of questions are answered correctly within one second (§VI). Surprisingly, the original is not the most accurate input for any of the six models. A compressed preview or thumbnail performs better in every case, with improvements of up to 15 percentage points. The difference is statistically significant for five of the six models (paired exact McNemar test, $p \leq 3 \times 10^{-4}$). For LLaVA-1.5-7B, JPEG Q10 and the original are tied within noise (75.6% vs. 74.7%, $p = 0.28$).

Always retrieve a thumbnail. A 32px thumbnail is inexpensive to transfer, but its usefulness varies considerably across models. Accuracy ranges from 20.2% for Pixtral-12B to 73.9% for Qwen2-VL-72B. A representation that works well for one model may perform poorly for another.

Escalate when the model is uncertain. A cascade can begin with a cheap representation and retrieve a more detailed one if the VLM reports low confidence [13]. This avoids some unnecessary transfers, but incurs additional retrieval and inference costs when escalation is needed. It can also stop prematurely when the model is confident despite missing visual evidence. For Pixtral-12B, such a cascade retrieves 1.36 MB per question and achieves only 44.3% accuracy (§VI-A, Table III).

Profile each model and fix its best representation. Choosing each model's most accurate representation on validation data avoids some of these problems. However, it still reads 0.86 MB per question on average (Table III), because the same representation is retrieved regardless of what the question asks. This is more than the bold entries in Table I suggest: those are chosen in hindsight, whereas a validation-based choice sometimes selects the original (for LLaVA-1.5-7B and Qwen2.5-VL-7B). Simple questions incur the cost of representations needed for more demanding ones.

These results suggest that neither a universal representation nor a fixed choice per model is sufficient. A better approach would select the representation for each question, taking into account both its expected accuracy and retrieval cost. The challenge is to make that decision before accessing the image.

C. What the workload reveals

We examine how answer accuracy varies across representations, models, and question types. Three observations motivate our approach.

Question form predicts the value of visual detail. Figure 1 (left) groups FloodNet questions by their surface form, without examining the underlying images. Averaged across six VLMs, counting questions beginning with *how many* benefit from increasing visual detail. Condition-recognition questions, in contrast, achieve their highest accuracy with a compressed JPEG preview, and some can be answered using captions or thumbnails. We observe similar differences on general VQA workloads. On VQA-v2, yes/no and color questions saturate at JPEG Q10, whereas counting and text-reading questions benefit from the original image.

The important distinction is that these patterns are associated with the question itself. They provide a signal for representation selection that is available before the image is retrieved.

More bytes do not necessarily produce better answers. Figure 1 (right) shows that accuracy is not monotone in representation size. Larger inputs can perform worse than smaller ones, and the preferred representation differs across VLMs. Representation selection therefore cannot be reduced to choosing the smallest input that approximates the original. It must account for the probability that a particular model will answer correctly using a particular representation.

Post-hoc confidence is an unreliable selection signal. Cascades typically decide whether to retrieve more detail after processing an initial representation. By then, the system has already incurred retrieval and inference costs. Moreover, model confidence does not reliably indicate whether the available visual evidence is sufficient. On VQA-v2, 43% of incorrect answers generated from captions have higher log-probability than the median correct answer. For Qwen2-VL-72B, log-probability predicts the need for escalation worse than random (AUC 0.338). We evaluate cascades in §VI-A and §VI-E.

Together, these observations point to a different approach: predict the usefulness of each representation directly from the question, rather than inspecting the image or relying on confidence after inference. Because these predictions are model-dependent and uncertain, they must be calibrated before they can support cost-based decisions.

D. The gap in existing systems

TABLE II

CAPABILITIES NEEDED BY THE USE CASE. R1: DECIDES BEFORE READING IMAGE DATA (METADATA AND INDEXES MAY BE READ). R2: AD HOC NATURAL-LANGUAGE QUERIES. R3: MULTIPLE REPRESENTATIONS OF AN ITEM. R4: MULTIPLE STORAGE TIERS. R5: MODEL-SPECIFIC ACCURACY ESTIMATES. R6: MULTI-OBJECTIVE COSTS AND WORKLOAD BUDGETS. ✓: SUPPORTED. ○: PARTIALLY (§II-D). ✗: NOT SUPPORTED. VOILA SUPPORTS R2 ONLY FOR QUESTION TEMPLATES SEEN DURING PROFILING (§VI-E).

System class	R1	R2	R3	R4	R5	R6
VLM APIs and serving [14]	✗	✓	✗	✗	✗	✗
Multimodal RAG / vector stores [5], [6]	○	✓	✗	✗	✗	✗
Visual storage managers [15]–[17]	✓	✗	✓	○	✗	○
ML-inference optimizers [18]–[22]	✗	✗	○	✗	✓	✓
LLM semantic query engines [1]–[4]	✗	✓	✗	✗	○	✓
Model routers and cascades [13], [23], [24]	✗	✓	✗	✗	✓	○
Storage tiering [25]	✓	✗	✗	✓	✗	○
VOILA	✓	○	✓	✓	✓	✓

Several lines of work address parts of this problem, but none jointly considers question-dependent representation accuracy and the cost of accessing representations across storage tiers. Table II summarizes the capabilities of existing system classes.

VLM APIs and serving systems process the inputs provided by the client. They do not generally decide which stored

representation should be retrieved. *Multimodal RAG systems and vector stores* focus on identifying which images or documents are relevant to a query. They decide which item to fetch from an index before reading it, but not which form of the item to fetch. They typically retrieve each selected item in a single stored representation, rather than choosing among alternative representations based on the question.

Visual storage managers such as VStore, VSS, and TASM [15]–[17] are closer to our setting. They maintain multiple physical formats or tiles and select inexpensive representations to serve requests. They weigh storage footprint against retrieval cost, but within a single store rather than across tiers, and without workload budgets. However, they generally assume that the consumer specifies its requirements, such as resolution, codec, or region, or that the downstream vision pipeline is fixed and can be profiled in advance. Natural-language questions do not specify these requirements explicitly, and the visual detail needed to answer them depends on both the question and the VLM.

ML-inference query optimizers [18]–[22] use proxy models, sampling, and input-resolution choices to reduce inference costs. Smol [22], for example, exploits the lower cost of processing small inputs, which is a partial form of representation choice. These approaches generally assume that the underlying visual data is accessible. Their decisions may process image content with inexpensive models, which still reads the data.

LLM semantic query engines [1]–[4] optimize model choice and operator execution against accuracy and cost targets, but typically treat an image column as a single stored value. Their accuracy estimates are per operator and model, not per input representation. *Model routers and cascades* [13], [23], [24] primarily choose among models or decide whether to invoke a more expensive model. They trade accuracy against model cost, but not under shared workload budgets. They do not address the joint choice of image representation and storage tier, and cascades incur additional retrieval costs when they escalate. Finally, *storage tiering systems* [25] place whole objects according to access patterns and storage price rather than the visual evidence required by individual queries.

An application could combine these techniques, but doing so requires substantial additional machinery. It would need to generate and catalog image representations, track their storage locations and access costs, profile VLM accuracy across representations, and predict which inputs are likely to answer each question correctly. It would also need to calibrate those predictions, enforce workload budgets and deadlines, and account for changes in model behavior or question distributions.

The missing component is a query layer that brings these decisions together. This is the role of VOILA: it uses question-derived estimates of answer correctness and cataloged access costs to select representations before image retrieval, while supporting both individual queries and workload-level optimization.

III. Data Model and Problem Formulation

The motivating workload suggests that the representation needed to answer a question can often be predicted before retrieving the image. We formalize the assets, costs, and correctness estimates used to make this choice, then define three optimization problems supported by VOILA.

A. Multi-representation, multi-tier assets

An image is an *asset* x with a set of available representations $\mathcal{R}(x) \subseteq \mathcal{R}$. Our deployment uses five representations:

$$\mathcal{R} = \{\textit{caption}, \textit{thumb}_{32}, \textit{jpeg}_{q1}, \textit{jpeg}_{q10}, \textit{original}\}.$$

A materialized instance (x, r) has size $b_r(x)$ and resides at $\ell(x, r)$ in the tier set

$$\mathcal{T} = \{\textit{device}, \textit{edge}, \textit{cloud}, \textit{archive}\}.$$

Representations are derived from the original at ingestion. A *representation catalog* records each instance's location, size, and storage pointer, and a pointer to caption text where applicable. Planning reads catalog metadata, not images.

B. Tiers and cost vectors

Retrieving representation r of asset x from tier t has cost vector

$$\mathbf{c}(x, r) = \big(b_r(x),\ L_t(b_r(x)),\ C_t(b_r(x)),\ I_M(r)\big).$$

Its components are transferred bytes, access latency, monetary cost, and VLM inference time. We model access latency as $L_t(b) = \alpha_t + b/\beta_t$, combining first-byte latency and transfer time. Monetary cost is $C_t(b) = \kappa_t + \pi_t b$, combining per-request and per-byte charges. Inference time $I_M(r)$ also depends on the representation: a caption is text, whereas an image produces visual tokens. A weight vector $\mathbf{w}$ combines these costs into a scalar objective. Applications may additionally impose hard constraints, such as a retrieval deadline or spending limit.

C. Queries and correctness

A query $Q = (q, x, M, \mathbf{w}, D)$ asks question q about asset x using VLM M, under weights $\mathbf{w}$ and deadline D. Let $Y_r(q, x) \in \{0, 1\}$ indicate whether M answers correctly using representation r. This outcome is not known when the optimizer makes its decision. Instead, it estimates

$$p_r(q) = \Pr\big[Y_r(q, x) = 1 \mid \phi(q)\big], \tag{1}$$

where $\phi(q)$ contains features extracted from the question and, optionally, precomputed catalog statistics. The estimator does not read a representation of x. The probability describes expected success over the workload, not a guarantee for an individual image.

D. Three optimization problems

Definition 1 (P1: Per-query selection). *Given Q and estimates $\widehat{p}_r$, choose*

$$r^*(Q) = \arg \max_{r \in \mathcal{R}(x):\, L_{\ell(x,r)}(b_r(x)) \le D} \left[\widehat{p}_r(q) - \lambda\, \mathbf{w}^\top \mathbf{c}(x, r)\right]. \tag{2}$$

The objective values a correct answer at one unit and trades expected correctness against weighted access and inference costs. Selecting r incurs the cost of that representation alone, rather than the cumulative cost of all smaller representations. This distinguishes direct selection from a cascade.

Definition 2 (P2: Workload allocation). *Given a batch $\mathcal{Q}$ and a budget B on scalar cost $c = \mathbf{w}^\top \mathbf{c}$, choose one representation $r(Q)$ for each query to maximize $\sum_{Q \in \mathcal{Q}} \widehat{p}_{r(Q)}$, subject to $\sum_{Q \in \mathcal{Q}} c(x_Q, r(Q)) \le B$.*

Definition 3 (P3: Representation placement). *Given a workload distribution $\mathcal{W}$, tier capacities K_t, and per-tier cost models, choose which asset–representation instances to materialize on each tier. The placement maximizes expected P1 utility for $\mathcal{W}$ subject to the storage capacity of each tier.*

P1 handles queries as they arrive. **P2** allocates a shared budget across a known batch, as when a response mission has a fixed backhaul allowance. **P3** is an offline physical-design problem, analogous to materialized-view and index selection [26], [27]. We do not solve P3 optimally; §IV gives a greedy placement advisor, which we evaluate in a two-tier simulation. The same correctness estimates used at query time also score candidate placements.

E. Guarantees

The quality of P1 depends on estimation error. The following bound makes this dependence explicit.

Proposition 1 (Selection regret)**.** *For query Q, let $\epsilon(Q) = \max_r |\widehat{p}_r(q) - p_r(q)|$ over feasible representations. Let r^* maximize the true objective in* (2)*, and let $\widehat{r}$ maximize the estimated objective over the same feasible set. Then the utility lost by choosing $\widehat{r}$ is at most $2\epsilon(Q)$. The expected regret is therefore at most $2\,\mathbb{E}_Q[\epsilon(Q)]$.*

Proof. Add and subtract the estimated utilities of r^* and $\widehat{r}$. The difference between their estimated utilities is non-positive because $\widehat{r}$ maximizes that objective. Each of the remaining two terms is bounded by $\epsilon(Q)$; the cost terms cancel because the same costs are used in both objectives. Taking expectations gives the second claim. □

The guarantee relates selection regret to the estimator's error on the operating workload. It does not bound error uniformly across unseen query distributions or establish a finite-sample guarantee for calibration. We therefore use disjoint populations for estimator training, calibration, and operating-point selection (§IV).

P2 is a multiple-choice knapsack problem (MCKP). Each query is a class and each representation is an item, with value $\widehat{p}_r$ and cost c. Although MCKP is NP-hard, its LP relaxation can be solved by considering incremental value per cost on the upper convex hull of each class. An optimal basic LP solution has at most one fractional class [28]. Provided the cheapest allocation is feasible, rounding that class down loses at most one expected correct answer relative to the integer optimum, since $0 \le \widehat{p}_r \le 1$. This gives a useful bound for large batches.

IV. VOILA Design

Figure 2 shows the online and offline components of VOILA. At query time, the system extracts text features, estimates the probability of a correct answer for each available representation, and combines those estimates with cataloged access costs. The executor then retrieves the chosen representation and invokes the target VLM once. Separately, ingestion creates representations, profiling trains and calibrates the estimators, and a placement advisor determines which tiers should hold them.

A. Query interface and catalog

VOILA exposes representation selection as an access-path choice for a VLM operator. Consider a semantic query:

```
SELECT a.id, ASK(a.image, 'How many buildings
  are flooded?', 'qwen3-vl') FROM aerial a
WHERE a.sector = 4
WITH (accuracy_floor = 0.95, deadline = 1s)
```

Here `a.image` denotes an asset, rather than its bytes. The query planner leaves the representation passed to `ASK` unresolved. For each qualifying asset, VOILA uses the question and catalog metadata to choose a representation before retrieving the image. The executor can group these reads by tier, while P2 can allocate a shared budget across the qualifying queries. This decision can also be inserted into an existing semantic query engine [1], [2]: VOILA chooses the input representation, leaving operator ordering and model selection to the host engine.

The representation catalog has schema

```
Rep(asset_id, rep, tier, uri, bytes,
  derived_from, created_at, caption_text_uri)
```

and stores one row per materialized instance. Tier parameters $(\alpha_t, \beta_t, \kappa_t, \pi_t)$ are maintained separately. Changes in prices or measured bandwidth update cost estimates without retraining the success estimators. Each target VLM also has a profile containing its representation estimators, calibrators, validated operating point, and label provenance. Since planning requires only metadata and model profiles, the catalog can be replicated at the edge or on devices.

B. Ingestion and materialization

The materializer generates representations where the original is first available. It creates thumbnails and JPEG previews by re-encoding the image, and uses a captioning model to generate a caption. Captioning is performed once per asset, so its creation cost is amortized over subsequent queries. Our online cost accounting excludes that one-time cost, which we did not measure; we also evaluate VOILA without captions (§VI-E). The materializer records the size and location of each resulting representation in the catalog.

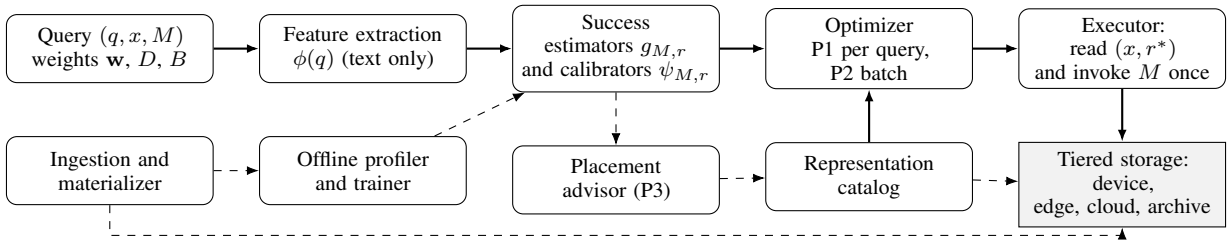


Fig. 2. VOILA architecture. The solid path performs query-time selection using the question and catalog metadata; only the executor reads a representation. Dashed arrows show offline ingestion, profiling, and placement.

C. Offline profiling

For each target VLM M, profiling runs the model on labeled questions at every representation and records the correctness outcome Y_r. This is the only stage in the design that deliberately evaluates multiple representations for the same question. The resulting labels are used to train the success estimators. Each outer fold's training questions are split by group into 70% for the estimators, 15% for calibration, and 15% for operating-point selection; on FloodNet this leaves 1,156 labeled questions per fold. Fitting all five estimators and calibrators takes 0.5 s on FloodNet and 41 s on 12,000 VQA-v2 questions on one laptop CPU (Apple M4).

D. Success estimators

The input to each estimator is a feature vector $\phi(q)$ built from the question text. It contains sparse TF–IDF features and structural indicators, including counting expressions, yes/no auxiliaries, numerals, negation, superlatives, and question length. For each model–representation pair (M, r), a gradient-boosted estimator $g_{M,r}$ predicts a raw success score. The estimators learn from the model's observed correctness labels; they do not inspect image content at query time.

The estimator family is replaceable. With calibration and the optimizer held fixed, gradient boosting, logistic regression, ridge regression, and an MLP differ by at most 1.6 accuracy points for Pixtral-12B on VQA-v2, and a fine-tuned BERT regressor in the same pipeline does not consistently improve the accuracy–cost trade-off (both in §VI-E). We use gradient boosting because it has low training and prediction overhead in this setting. The interface also permits features computed at ingestion and recorded in the catalog, such as representation sizes. We do not rely on such features for the reported question-text-only results.

E. Per-representation calibration

Raw estimator scores are not directly comparable: a score of 0.6 for a caption need not imply the same success probability as 0.6 for the original. Because the optimizer trades estimated correctness against cost, these scores must be calibrated. For each (M, r), VOILA fits an isotonic map $\psi_{M,r}$ on a population disjoint from estimator training (the 15% calibration split), producing

$$\widehat{p}_r(q) = \psi_{M,r}\big(g_{M,r}(\phi(q))\big).$$

Algorithm 1 Per-query representation selection (P1)

Require: query (q, x, M), weights $\mathbf{w}$, deadline D, penalty λ
$F \leftarrow \{r \in \mathcal{R}(x) : L_{\ell(x,r)}(b_r(x)) \le D\}$ ▷ catalog only
if $F = \emptyset$ **then**
  $F \leftarrow \{\arg\min_{r \in \mathcal{R}(x)} L_{\ell(x,r)}(b_r(x))\}$ ▷ best effort; deadline unmet
end if
for all $r \in F$ **do**
  $\widehat{p}_r \leftarrow \psi_{M,r}(g_{M,r}(\phi(q)))$
  $u_r \leftarrow \widehat{p}_r - \lambda\, \mathbf{w}^\top \mathbf{c}(x, r)$
end for
$r^* \leftarrow \arg\max_{r \in F} u_r$
Read (x, r^*) from $\ell(x, r^*)$; **return** $M(q, (x, r^*))$

We calibrate each representation separately and do not require correctness to increase with representation size; Table I shows why such a constraint would be inappropriate.

F. Per-query optimizer

Algorithm 1 implements P1. Using catalog metadata, it excludes representations that are unavailable for the asset or violate the retrieval deadline. It evaluates the lightweight estimators for the remaining candidates, computes their expected utilities, and selects the maximizer. The online path reads one representation and invokes the VLM once, rather than retrieving progressively larger inputs. Planning a query takes 0.64 ms (median; 7 μs per query when batched) on one laptop CPU core, which is small next to the access latencies of §VI-B.

The operating-point parameter λ is selected on validation data, separate from both training and calibration. Let $A^{\text{val}}_{\text{fix}}$ be the accuracy of the best fixed representation on that population. Among candidate penalties achieving at least $0.95A^{\text{val}}_{\text{fix}}$, VOILA selects the one that transfers the fewest bytes. This is a relative validation accuracy target, not a per-query guarantee; applications may choose a stricter trade-off.

Unlike sequential escalation, P1 compares the full utility of each available representation. An earlier incremental-gain heuristic remains in the experimental ablations, but does not implement the P1 objective.

G. Workload optimizer

For a known batch with a shared budget, VOILA solves P2 as an MCKP: each query contributes one class, with a

candidate item for each representation. We implement two allocators. *Bucketed dynamic programming* rounds costs up to 1 KiB buckets and retains Pareto-optimal (cost, value) states. Rounding up makes the returned allocation feasible under the unrounded budget, though it need not be optimal for the original continuous-cost instance. In our measurements, a 100-query batch is solved in under a second. *Marginal-value allocation* is a greedy alternative: it initially assigns each query its cheapest representation, then applies upgrades in decreasing order of predicted accuracy gain per additional unit cost, provided they fit within the remaining budget.

Both methods plan over a batch; single queries use P1.

H. Placement advisor

P3 uses the same correctness estimates to guide representation placement. For a candidate placement, the placement advisor replays a sample of the workload through the P1 decision rule, restricted to representations available at the modeled tiers. It measures the expected utility under that placement. Starting from representations stored in the cloud, it greedily copies the asset–representation instance with the largest popularity-weighted utility gain per added storage byte to a faster tier, stopping at the tier's capacity. Popularity weights come from the expected workload. The cloud copy is retained, and P1 reads the fastest available copy. A copy can improve utility by lowering a tier-dependent cost or by making a representation feasible under a deadline. With the byte objective we evaluate, cost does not depend on the tier, so all of the placement advisor's measured gains come from deadline feasibility. Placement evaluation uses the success estimators rather than invoking VLMs, so the placement advisor scores every (image, representation) candidate, 7,225 on FloodNet, offline (§VI-D).

I. Maintenance

New representations. Since estimators are defined per model–representation pair, adding a representation requires profiling and training its estimator and calibrator; existing representation estimators do not need to be retrained.

New models. Estimators trained on an earlier model can provide a starting point for a newer one, with recalibration as new labels become available. Within the Qwen family, zero-shot transfer is 3.4–4.5 accuracy points below a target-trained estimator on VQA-v2 and matches its accuracy on GQA (§VI-E).

Unseen question forms. Question-text estimators are less reliable on templates absent during profiling (§VI-E). For workloads drawn from a fixed set of question templates, as in FloodNet, the model profile records the normalized question strings (lowercase, punctuation removed) seen during profiling. A query whose string is not recorded reads the best fixed representation on the profiling data until labels for its template are collected. Because matching is exact, a rephrased profiled question is also treated as new. That forgoes savings on the query but avoids the held-out template failure. A feature-distance test does not work: every held-out FloodNet template has TF–IDF cosine similarity of at least 0.8 to a profiled one (§VI-E). Exact matching does not suit open-ended workloads. On VQA-v2, GQA, and TextVQA, 69–82% of test questions have a string never seen in training, so the fallback would disable selection for most queries. We do not use it on those workloads.

V. Experimental Setup

Primary workload. We use the FloodNet visual question answering benchmark [12] as the disaster-response workload. The common population contains 1,445 questions. For each, all five representations and the outputs of all six VLMs are available. We report measured payload bytes per representation (Table I). Each of five outer folds holds out images, not questions. The remaining images are split into disjoint populations for estimator training (70%), calibration (15%), and operating-point validation (15%). Questions about the same image therefore never cross a boundary. Confidence intervals and paired differences resample image clusters, with 2,000 bootstrap replicates.

Breadth workloads. To test generality, we also use VQA-v2, GQA, and TextVQA [29]–[31]. These runs use a tier-weighted cost, normalized so that reading the original costs 100. The other costs are 7.6 (caption), 15.1 (thumbnail), 37.8 (JPEG Q1), and 53.2 (JPEG Q10). They are derived from measured size ratios and a base cost per tier (edge, warm, cold). These runs use question-level five-fold cross-validation. We also report an image-grouped re-run.

Models. We use Pixtral-12B [32], Llama-4-Mav. [33], LLaVA-1.5-7B [34], Qwen2-VL-72B [35], Qwen2.5-VL-7B [36], and Qwen3-VL-235B [37]. All run zero-shot. They span 7B to 235B parameters.

Baselines. *Fixed* always reads one representation. *Best fixed* is the fixed representation with the highest validation accuracy in each fold. *Log-prob cascade* is a FrugalGPT-style cascade [13]. It reads representations in increasing size. It stops when the VLM's answer log-probability exceeds a validation-tuned threshold. Its cost is cumulative. *BERT router* is a supervised sentence-embedding classifier that predicts the cheapest correct representation. *GPT-4o router* (breadth only) has GPT-4o label each question's visual need as low, medium, or high. *Oracle* picks the cheapest representation that answers correctly, in hindsight.

Metrics. We report accuracy and mean bytes per query (FloodNet) or normalized cost (breadth). Under recorded network traces we report deadline-qualified accuracy (DQA), the probability that an answer is correct and its read finishes within deadline D: $\mathrm{DQA}(D) = \Pr[Y = 1 \wedge T_{\mathrm{access}} \leq D]$. For workloads, we report accuracy under a shared batch budget.

VI. Evaluation

We ask five questions. (Q1) Does query-driven selection improve the accuracy–cost frontier over fixed and adaptive policies? (Q2) Do the gains hold under other cost objectives? (Q3) How should a workload budget be allocated? (Q4) What do selections imply for placement? (Q5) How robust and maintainable is VOILA, and where does it fail?

TABLE III
FLOODNET ACCURACY (%) AND MEASURED MB PER QUERY. BEST FIXED IS CHOSEN ON VALIDATION DATA. CASCADE BYTES ARE CUMULATIVE.

	Original		Best fixed		Cascade		BERT		VOILA	
Model	Acc	MB	Acc	MB	Acc	MB	Acc	MB	Acc	MB
Llama-4-Mav.	63.1	5.52	69.2	0.400	52.4	0.120	72.5	1.000	68.2	0.087
LLaVA-1.5-7B	74.7	5.52	74.9	2.386	73.1	0.047	76.5	0.903	75.7	0.015
Pixtral-12B	38.8	5.52	50.5	0.445	44.3	1.364	53.0	1.384	54.3	0.242
Qwen3-VL-235B	76.6	5.52	83.2	0.445	67.3	0.052	82.5	1.120	82.1	0.113
Qwen2-VL-72B	58.8	5.52	73.9	0.002	70.3	0.910	40.9	0.756	73.3	0.002
Qwen2.5-VL-7B	67.4	5.52	70.9	1.461	60.1	0.487	70.3	1.641	69.2	0.358
Mean	63.2	5.52	70.4	0.857	61.3	0.497	66.0	1.134	70.5	0.136

A. Q1: Per-query selection

VOILA reaches best-fixed accuracy at a fraction of the bytes. Table III and Figure 3 show the results. VOILA averages 70.5% accuracy at 0.136 MB per query. The best fixed representation averages 70.4% at 0.857 MB, and it requires knowing in advance which representation suits each model. Reading originals averages 63.2% at 5.52 MB. VOILA is thus as accurate as the best fixed policy while reading 84% fewer bytes. It is 7.3 points more accurate than reading originals while reading 97.5% fewer bytes.

The gain is model-dependent. We compare VOILA with best fixed using paired, image-clustered resampling. VOILA is more accurate by 3.8 points on Pixtral-12B and by 0.8 points on LLaVA-1.5-7B. It is less accurate by 0.6–1.7 points on the other four models. It reads fewer bytes in every case. For Qwen2-VL-72B, best fixed is the 32-pixel thumbnail. VOILA reads 1,916 bytes per query against the thumbnail's 2,141 and is 0.6 points less accurate. Neither policy dominates. The 95% confidence intervals on VOILA's accuracy span 4–5 points per model. Those on its bytes are narrow (e.g., Qwen3-VL-235B: 0.103–0.122 MB). The byte savings are therefore not driven by a few large originals.

Supervised routing is more accurate on some models but reads far more. The BERT router is 4.2 points more accurate than VOILA on Llama-4-Mav.. It reads 1.13 MB per query on average, 8× more than VOILA. On Qwen2-VL-72B it collapses to 40.9%. It predicts a label (the cheapest correct representation) rather than comparable probabilities. It therefore has no way to trade accuracy against cost.

Cascades pay for confidence they cannot trust. The log-prob cascade is cheaper than VOILA on some models. It sacrifices up to 14.8 points in doing so (Qwen3-VL-235B: 67.3% vs. 82.1%). On Pixtral-12B it reads 1.36 MB per query and reaches 44.3%, worse than VOILA on both axes.

The same holds across datasets. Table IV reports the breadth workloads under tier-weighted cost. VOILA reduces cost by 65–73% relative to originals. On TextVQA it is within 1.8 points of the original at 27% of its cost. On VQA-v2 and GQA, the accuracy gap to the original is 10.1 and 10.6 points. A fixed JPEG Q10 preview is more accurate than VOILA on these two datasets, but it costs 51–91% more. VOILA dominates the log-prob cascade on all three datasets. It dominates the BERT router on VQA-v2 and TextVQA. On GQA it ties the BERT router (59.6% vs. 59.7%) at 32% lower cost. The GPT-4o router is more accurate on VQA-v2. It costs 40% more and requires a large-model call per query.

TABLE IV
BREADTH WORKLOADS: MEAN ACCURACY (%) / NORMALIZED TIER-WEIGHTED COST (ORIGINAL = 100) OVER SIX VLMS. JPEG Q10 IS THE STRONGEST FIXED PREVIEW.

Policy	VQA-v2	GQA	TextVQA
Caption	47.8 / 7.6	39.9 / 7.6	75.7 / 7.6
JPEG Q10	72.0 / 53.2	67.0 / 53.2	81.3 / 53.2
Original	75.0 / 100	70.2 / 100	86.6 / 100
Log-prob cascade	54.5 / 36.6	51.1 / 44.7	71.9 / 39.3
GPT-4o router	67.2 / 49.2	57.7 / 40.5	84.5 / 70.7
BERT router	62.0 / 40.9	59.7 / 40.9	71.9 / 33.1
VOILA	64.9 / 35.2	59.6 / 27.9	84.8 / 26.6
Oracle (hindsight)	86.9 / 29.8	83.5 / 33.2	96.7 / 18.2

TABLE V
RECORDED-TRACE ACCESS LATENCY (S) AND DEADLINE-QUALIFIED ACCURACY (%), AVERAGED OVER SIX VLMS AND SIX UPLINK SCENARIOS. C@1: FRACTION OF READS COMPLETING WITHIN 1 S.

Policy	MB	p50	p95	C@1	DQA@1	DQA@10
Original	5.521	1.428	2.959	59.3	37.9	63.0
Best fixed	0.359	0.112	0.281	99.5	70.3	70.8
Cascade	0.497	0.034	1.237	96.3	59.7	61.2
BERT router	1.134	0.054	2.370	91.7	63.6	65.9
VOILA	0.136	0.052	0.230	99.7	70.3	70.5

B. Q2: Cost objectives beyond bytes

Latency under recorded networks. We replay six public CellReplay uplink traces [38]. They cover two carriers and driving, stationary, weak-signal, crowded, and walking scenarios. We treat each recorded delivery opportunity as a 1,400-byte slot and sample 100 start phases per query. Table V reports serialization latency and deadline-qualified accuracy (DQA, §V), averaged over models and scenarios. Reading originals completes within one second for 59.3% of queries. Its one-second DQA is 37.9%. VOILA completes 99.7% of reads within one second and reaches 70.3% DQA. Its p95 access latency is 0.23 s, versus 2.96 s for originals. A deadline-aware best fixed policy matches VOILA's DQA. It reads 2.6× the bytes and has a higher p95 latency. The DQA gap to originals persists in every scenario. In the weak-signal trace, no original arrives within one second. VOILA answers 69.8% of questions correctly in time.

Tier-weighted access cost. The breadth results in Table IV already price representations by tier, not by bytes alone. Captions and thumbnails are served from an edge tier. Previews come from a warm tier, and originals from a cold tier. P1 depends only on cost differences. Rescaling or shifting costs while preserving their order therefore leaves the per-query decision unchanged for a fixed cost weight λ (Eq. 2). Only the operating point moves.

Inference cost. Representation choice also changes VLM inference time. This addresses the concern that acquisition is only part of the cost. We measured inference on 4×V100 GPUs.

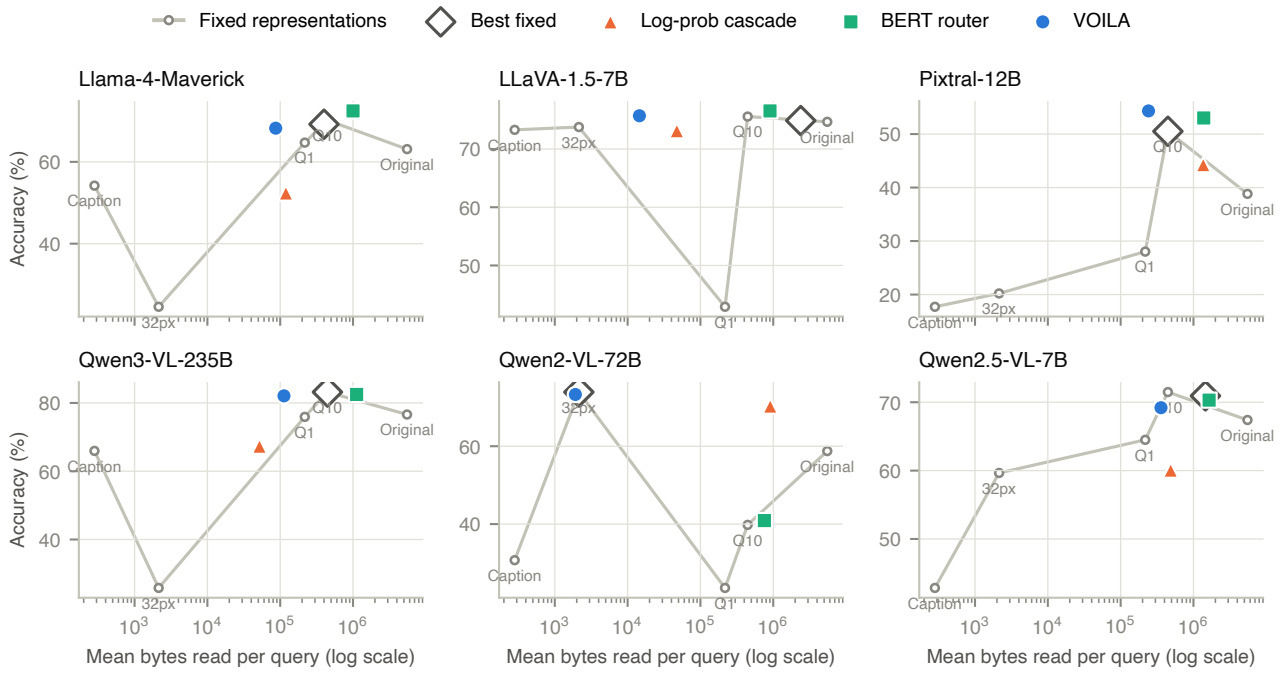


Fig. 3. Accuracy versus measured bytes on FloodNet. Every point uses the same image-grouped test population. Fixed points trace the five representations. Cascade points include every representation read before stopping.

A caption-based query takes 2.4–12.5 s across the six models. The same query with the original takes 5.7–62.8 s. For Qwen3-VL-235B, answering from a caption is 5× faster than from the original (12.5 s vs. 62.8 s). We did not measure inference on the 32px and JPEG Q1 representations, so we do not evaluate expected inference time per policy. All reported objectives set the inference weight to zero. *The reported savings are therefore conservative* whenever VOILA avoids the original.

C. Q3: Workload allocation under budgets

We allocate a shared byte budget over batches of 100 FloodNet questions, the workload with measured per-asset bytes (Table VI). The budget is a fraction of the bytes needed to read every original (552 MB per batch). We draw 100 random batches per setting and average over the six models. Both allocators beat the best fixed representation that fits the budget. They gain 1.6 points at 10% and 3.1 points at 50%. The gain is positive in 17 of 18 model–budget settings and ranges from 1.0 to 7.0 points. The exception is Qwen2.5-VL-7B at 10% (−0.6). Averaged over models, exact and marginal-value allocation are within 0.1 points at every budget. On this workload the greedy relaxation loses nothing. The hindsight oracle reaches 85.1%, so the remaining headroom lies in estimator quality, not in the allocator. The legacy sequential rule ignores the budget and stays at 53.2%.

Allocators that maximize $\sum \hat{p}$ spend most of the budget. At 50%, they read 133 MB per batch, against 36 MB for the best fixed representation. A deployment that values unspent budget should add a per-byte penalty to item values, as P1 does.

TABLE VI
FLOODNET ACCURACY (%) FOR 100-QUESTION BATCHES UNDER A BYTE BUDGET (FRACTION OF READING ALL ORIGINALS), AVERAGED OVER SIX VLMS AND 100 RANDOM BATCHES.

Policy	10%	25%	50%
Best feasible fixed	70.6	70.6	70.6
Exact allocation (DP)	72.2	73.3	73.7
Marginal-value allocation	72.1	73.3	73.7
Sequential (legacy) rule	53.2	53.2	53.2
Oracle (hindsight)	85.1	85.2	85.2

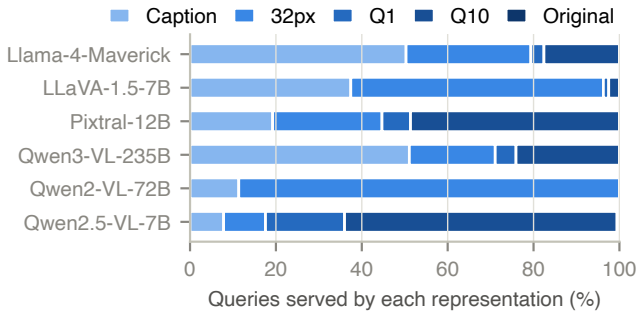


Fig. 4. Representations selected by VOILA on FloodNet, by model. Every model sees the same questions; the mix reflects its measured behavior.

D. Q4: Representation placement

Figure 4 shows which representations VOILA reads. The caption and the 32-pixel thumbnail together serve 100% of Qwen2-VL-72B's queries. They serve 96% of LLaVA-1.5-7B's,

79% of Llama-4-Mav.'s, and 71% of Qwen3-VL-235B's. They serve only 45% of Pixtral-12B's and 18% of Qwen2.5-VL-7B's queries, which rely on JPEG Q10. For every model, the original is read for fewer than 1% of queries.

These mixtures have direct placement consequences. Captions and thumbnails together occupy about 2.4 KB per image, or 0.04% of the original's size. Materializing both at the edge for every asset therefore costs almost nothing. Averaged across models, it serves 68% of queries without crossing the backhaul. JPEG previews add 0.66 MB per image, or 12% of the original. They are worth placing at the edge only for models that select them, such as Pixtral-12B and Qwen2.5-VL-7B. Originals are read for under 1% of queries. They can move to a cold or archive tier with little effect on accuracy. Placement thus depends on the model mix, so VOILA optimizes it with its own estimates rather than a fixed policy.

Placement under capacity sweeps. We evaluate the placement advisor in a two-tier setting. All representations are stored in the cloud. An edge tier holds a fraction of the corpus (7.98 GB of originals). Cloud reads follow the recorded uplinks of Table V. For each scenario we fit the p95 read latency as $\alpha+b/\beta$ ($R^2 \geq 0.945$). Edge reads take 2 ms plus $b/(100\,\text{MB/s})$, an assumed LAN-attached edge server. Image popularity is Zipf(1), and we draw ten random popularity orders. Every query is routed by Algorithm 1 at its validated λ, with a 0.25 s read deadline. A read that misses the deadline counts as wrong. We compare four placements (Table VII). *Frequency tiering* fills the edge with the originals of the most popular images, the usual hot/cold policy. We pair it either with reading originals, as today's pipelines do, or with VOILA's routing. *Size-ordered classes* places whole representation classes in ascending size. The *placement advisor* (§IV-H) fills the edge greedily with the (image, representation) copies of largest popularity-weighted accuracy gain per byte. Routing cost is bytes, which are the same on either tier, so an edge copy helps only when the cloud copy would miss the deadline.

Frequency tiering of originals answers at most 48.5% of questions within the deadline, even with 20% of the corpus at the edge. Representation selection matters more than placement. With no edge tier at all, VOILA's routing reaches 67.1%. Placement then closes the remaining gap. With 0.5% of the corpus at the edge, the placement advisor reaches 68.5%. Size-ordered classes need 5% to match it. Frequency tiering gains nothing, because VOILA rarely reads the originals it caches. The placement advisor's gains come from the two poorest uplinks. Under the weak-signal trace, 1% of edge capacity raises accuracy from 62.1% to 67.8%, and the placement advisor uses only 23 MB of the 80 MB allowed. Size-ordered classes reach 63.1%. On the four better uplinks, every placement reaches the deadline-free ceiling of 68.9%. With a 1 s deadline, placement does not matter for VOILA's routing, while frequency tiering of originals reaches at most 53.2%. Uniform popularity gives the same ordering.

TABLE VII
DEADLINE-QUALIFIED ACCURACY (%, 0.25 S READ DEADLINE) BY EDGE CAPACITY (% OF ORIGINAL BYTES). AVERAGED OVER SIX VLMS, SIX RECORDED UPLINKS, AND TEN ZIPF(1) POPULARITY ORDERS.

Placement	0%	0.5%	1%	5%	20%
Frequency tiering, read originals	0.0	19.1	24.2	37.4	48.5
Frequency tiering + VOILA	67.1	67.1	67.1	67.1	67.2
Size-ordered classes + VOILA	67.1	67.3	67.3	68.5	68.9
Placement advisor + VOILA	67.1	68.5	68.7	68.9	68.9

TABLE VIII
ROBUSTNESS ON FLOODNET (ACCURACY %, PER QUERY). TEMPLATE: ENTIRE QUESTION TEMPLATES HELD OUT OF TRAINING. +FB: UNPROFILED TEMPLATES ROUTED TO THE VALIDATED FALLBACK REPRESENTATION. STRUCT: TF–IDF REMOVED. CALIBRATION COLUMNS REPORT ACCURACY / MB PER QUERY.

Model	Main	Template	Δ	+FB	Struct	No calib.	Isotonic
Llama-4-Mav.	68.2	59.9	−8.4	61.9	68.4	67.8/0.076	68.2/0.086
LLaVA-1.5-7B	75.7	74.3	−1.4	74.4	76.1	75.8/0.001	75.7/0.015
Pixtral-12B	54.3	21.3	−33.0	50.5	53.8	54.8/0.224	54.3/0.232
Qwen3-VL-235B	82.1	33.2	−48.9	83.2	81.1	83.2/0.134	82.1/0.118
Qwen2-VL-72B	73.3	58.5	−14.7	73.9	73.4	74.0/0.002	73.3/0.002
Qwen2.5-VL-7B	69.2	63.3	−5.9	65.2	68.7	69.3/0.356	69.2/0.354

E. Q5: Robustness, maintenance, and failure

Unseen question templates are the dominant failure mode. Holding out entire question templates reduces per-query accuracy by 1.4–48.9 points, from 70.5% to 51.8% on average (Table VIII). The loss is concentrated. FloodNet's profiled workload contains 15 distinct question strings. One of them, "What is the overall condition of the given image?", accounts for 49% of queries. When it is held out, Qwen3-VL-235B routes it to the 32-pixel thumbnail and answers 5.7% correctly. When it is profiled, Qwen3-VL-235B answers it correctly 96.7% of the time. Image-grouped folds prevent visual leakage. They do not guarantee robustness to new phrasings.

The fallback recovers the loss. Routing unprofiled templates to the best fixed representation on the training folds raises template-holdout accuracy from 51.8% to 68.2%, matching that representation (Table VIII, +FB). Profiled templates are unaffected, so in-distribution accuracy is unchanged. Detecting novelty by feature distance does not work here. Every held-out template has TF–IDF cosine similarity of at least 0.8 to a profiled one. Any threshold that flags the failing template flags nearly every held-out query. Template novelty is the reliable signal, because templates recur in analytic workloads like prepared statements.

Image-grouped protocol on breadth workloads. We re-ran VQA-v2 and GQA with image-disjoint folds. Mean accuracy over six models changes from 64.9% to 63.7% on VQA-v2 and from 59.6% to 58.8% on GQA. Fold standard deviations rise from under 1 point to 2.9 and 5.8 points. Leakage through shared images or templates does not explain the results. Generalization to new templates is, however, less stable.

Cross-model transfer. An estimator trained on one model's labels can serve another model without retraining (Table IX).

TABLE IX
CROSS-MODEL TRANSFER (ACCURACY % @ NORMALIZED COST). ZERO-SHOT: ESTIMATOR TRAINED ON THE SOURCE MODEL'S LABELS.

Data	Source → target	Zero-shot	Target-trained	Δ
VQA-v2	Qwen2-72B → Qwen3-235B	67.9@30.8	71.5@37.4	−3.6
	Qwen2.5-7B → Qwen3-235B	67.0@30.9	71.5@37.4	−4.5
	Qwen3-235B → Qwen2-72B	61.0@35.6	64.4@37.3	−3.4
GQA	Qwen2-72B → Qwen3-235B	61.9@26.5	59.0@24.3	+2.9
	Qwen2.5-7B → Qwen3-235B	60.4@34.3	59.0@24.3	+1.4
	Qwen3-235B → Qwen2-72B	60.9@29.6	60.0@29.0	+0.9

On VQA-v2 it stays within 3.4–4.5 points of a target-trained estimator. On GQA it matches it. An existing estimator can therefore serve an updated model while new labels accumulate.

Simple structure carries much of the signal. Removing TF–IDF and keeping only structural features changes FloodNet accuracy by at most one point (Table VIII). Under shift between VQA-v2 and GQA, only 49–57% of the TF–IDF vocabulary overlaps. TF–IDF permutation importance drops by 62–100%. Structural features retain most of theirs. Lexical features are high-capacity but corpus-specific; structural ones transfer.

Calibration is a model-selection choice. On FloodNet, isotonic calibration does not uniformly improve the selected operating point. No calibration is slightly better for four models (Table VIII). On VQA-v2 and GQA, isotonic calibration reduced cost by 10–25% relative to uncalibrated routing at comparable accuracy. For Pixtral-12B on VQA-v2, it reduced expected calibration error from 0.025 to 0.012. On FloodNet, the out-of-fold isotonic estimates have expected calibration error 0.021–0.029 (ten bins, all five representations) and Brier score 0.098–0.170, against 0.214–0.248 for a constant predictor. Estimator training, calibration, operating-point selection, and testing use disjoint question groups in every fold. Calibration is required for P1's subtraction to be meaningful. Which calibrator works best should be selected on validation data.

Estimator family and fair baselines. We fix calibration and the optimizer and vary the estimator for Pixtral-12B. Gradient boosting, logistic regression, ridge, and MLP estimators are within 1.6 accuracy points and 5 cost units of each other on VQA-v2. A fine-tuned BERT regressor in the same pipeline reaches 66.0% at cost 42.6 on VQA-v2 and 60.1% at 35.2 on GQA. Gradient boosting reaches 64.1% at 38.8 and 60.5% at 31.1. BERT thus brings no consistent improvement, at a much higher estimator cost. Calibrating the cascade's log-probabilities does not close its gap either. On TextVQA, the calibrated cascade reaches 71.2% at cost 26.3, versus 84.8% at 26.6 for VOILA.

Objective consistency and the caption tier. Replacing the legacy sequential rule with the rule of (2) changes breadth results by +0.4 accuracy points and +0.7 cost units on average. On FloodNet, the legacy rule is markedly worse (e.g., 25.8% vs. 54.3% for Pixtral-12B). It under-reads when gains are non-monotone. Removing the caption tier keeps VOILA within 2.4 points on VQA-v2, GQA, and FloodNet. It costs 14.4 points on TextVQA. There, captions are generated from originals and often transcribe scene text. TextVQA results should therefore be read as depending on that derived view.

TABLE X
FAILURE DECOMPOSITION ON FLOODNET (% OF QUERIES). ORACLE: ANSWERABLE BY SOME REPRESENTATION. EXACT: VOILA PICKS THE CHEAPEST CORRECT ONE. UNDER: WRONG ALTHOUGH SOME REPRESENTATION WAS CORRECT. OVER: CORRECT, BUT WITH MORE BYTES THAN NEEDED. NONE: NO REPRESENTATION IS CORRECT.

Model	Oracle	Exact	Under	Over	None
Llama-4-Mav.	88.6	51.8	20.3	16.4	11.4
LLaVA-1.5-7B	83.7	33.6	8.0	42.1	16.3
Pixtral-12B	79.4	40.9	25.1	13.4	20.6
Qwen3-VL-235B	90.2	59.4	8.2	22.7	9.8
Qwen2-VL-72B	86.3	46.2	13.0	27.1	13.7
Qwen2.5-VL-7B	84.2	7.8	14.9	61.4	15.8

Where the remaining gap lies. Table X separates reducible from irreducible error. Under-reads affect 8–25% of queries and cost accuracy. Better estimates, for example from catalog statistics or fallbacks, can recover them. Over-reads affect 13–61% of queries and cost bytes. Sharper cost discrimination can recover them. Finally, 10–21% of queries are answered incorrectly at every representation. Selection cannot fix them.

Overhead. The saved estimator bundle for one model is 234,666 bytes. A decision takes 4.12 ms at the median and 5.32 ms at p95 on a reference CPU. Peak process memory is 162 MB. This is two to four orders of magnitude below VLM inference (2.4–62.8 s). It is also far below the access latency of an original over the recorded uplinks (p50 1.43 s).

VII. DISCUSSION

When does query-driven selection pay off? The gains are largest in two conditions. The workload mixes questions with different evidence needs, and a model's best fixed representation is large. For Llama-4-Mav. and Qwen3-VL-235B, the best fixed input is a 445 KB preview. VOILA serves most of their questions from a caption or thumbnail. It reads 75–80% fewer bytes at a 1-point accuracy cost. When a model's best fixed input is already tiny, little is left to save. This is the case for Qwen2-VL-72B, whose best input is a 2.1 KB thumbnail. There, the optimizer correctly converges to almost the same policy.

Estimates as optimizer statistics. VOILA's success estimators play the role that cardinality estimates play in a relational optimizer. They price alternatives that the optimizer never executes. Their errors turn into plan regret (Proposition 1). The same operational practices apply. Estimates must be refreshed when the workload drifts. Template holdout is the analogue of a stale histogram. Estimates must be validated on data the optimizer did not train on. They need a conservative fallback for queries they were never fitted to, as an optimizer falls back to a safe plan when statistics are missing.

Accuracy floors as service-level objectives. The validation rule keeps accuracy within 95% of the best fixed representation. This is a contract, not a constant. A medical archive would raise it. A best-effort photo search would lower it. Because the estimates are probabilities, changing the contract changes only the operating point. The estimators stay the same.

Composition. Representation selection composes with model routing, which chooses the model after the input is fixed. It also composes with in-model token reduction [39], [40], which shrinks the computation on the bytes that were read. A joint optimizer over (representation, model) pairs is a natural extension. It would require success estimates for every pair.

Data minimization. Reading a caption or thumbnail instead of an original means fewer bytes leave the storage tier. Images often contain incidental personal information. Selecting the least detailed sufficient representation is then a data-minimization measure as well as a cost optimization.

VIII. Related Work

Visual data management. Video and visual DBMSs store multiple physical representations and choose cheap ones to serve queries. VStore selects storage and consumption formats for video analytics pipelines [15]. VSS stores and reuses multiple physical video representations [16]. TASM stores tiles so that queries read only regions of interest [17]. LightDB optimizes declarative queries over VR video [41]. These systems choose among representations that satisfy an explicitly stated format or a profiled vision pipeline. VOILA addresses the case where the consumer is a general VLM answering a free-form question. The required representation must then be inferred from the question and the model.

ML inference in query processing. NoScope, BlazeIt, probabilistic predicates, and TASTI reduce the cost of neural predicates and aggregates over video and unstructured data [18]–[21]. They use cheap proxies, indexes, and sampling. Smol jointly optimizes preprocessing and inference, including low-resolution inputs [22]. LLM-powered engines such as LOTUS, Palimpzest, ThalamusDB, and CAESURA optimize semantic operators and model choices over multimodal data [1]–[4]. These systems decide after data is local, or by running models on it. VOILA decides which stored representation to read before reading. It can supply its choice to any of them.

Approximate query processing and physical design. BlinkDB maintains samples of different sizes. It selects one per query to meet error and latency bounds [42]. VOILA analogously maintains representations of different fidelity. It selects one per query against an estimated accuracy, with VLM correctness as the error model. Placing representations across tiers is a physical-design problem, like view and index selection [26], [27]. VOILA's estimates serve as its cost model.

Routing and cascades. FrugalML, FrugalGPT, RouteLLM, and hybrid routing select among models or APIs for a given input [13], [23], [24], [43], [44]. Early-exit methods shorten computation [45], [46]. These methods optimize how an available input is processed. They can follow VOILA's selection.

Query difficulty and value of information. Pre-retrieval query performance prediction estimates retrieval quality from query features alone [47]–[49]. VQA analyses show that question type predicts difficulty [50], [51]. Value-of-information and cost-sensitive acquisition decide whether evidence is worth its cost [52]–[54]. VOILA applies these ideas to the representations of an identified asset. Model-specific correctness is the value, and catalog-priced access is the cost. Adaptive resolution inside VLMs [39], [40], [55] reduces computation on an image already read. It is complementary.

IX. Limitations

The question is a partial signal. The same question can require different detail for different images, such as a large versus a distant object, and compound or conditional questions blur the signal. Unseen templates are the main risk (§VI-E). The fallback recovers best-fixed accuracy but forgoes VOILA's savings until the new template is profiled, and it needs recurring templates. On open-ended workloads (69–82% unseen strings), VOILA still cuts cost by 65–73% (Table IV), but guarding against harmful novelty there would need a semantic novelty signal. Conditioning the estimators on catalog features such as caption text, and conformal novelty tests, are natural next steps.

Discrete, profiled representations. VOILA chooses among materialized representations and needs correctness labels per model; a new representation or model requires profiling only for the new component. Online profiling from production feedback could replace the offline labeling step.

Measured costs on one workload. Measured bytes and network traces are reported on FloodNet; the breadth datasets use a tier-weighted cost model. The trace experiment simulates serialization over recorded delivery opportunities and omits transport-protocol dynamics.

Derived captions. Captions are generated from originals at ingestion, and their one-time cost is excluded from per-query accounting. On OCR-heavy data, they carry information that low-resolution images do not. Deciding which representations to materialize, not only where, extends P3.

English questions. Features and question-type rules target English. Multilingual queries require new profiling data.

X. Conclusion

A VLM query over image data does not have to read the original. VOILA treats the choice of representation as an optimizer decision taken before any pixel is read. It uses calibrated, model-specific estimates of answer correctness, derived from the question text. It prices representations with a catalog of where each one lives and what it costs. On a disaster-response workload, it is as accurate on average as the best fixed representation chosen on validation data while reading 84% fewer bytes, and it nearly doubles the share of questions answered correctly within one second. The same estimates allocate shared workload budgets and, in a two-tier simulation, guide which representations an edge tier should hold. The main failure case is question templates absent from profiling. A template fallback contains it on templated workloads. Open-ended workloads mark where data systems will need richer evidence about what a query needs, still without reading pixels.

Acknowledgement of AI-Generated Content

Claude Code and Codex were used for grammar and editorial assistance in Sections I–X of this paper. They also assisted

with writing and revising the VOILA implementation and experimental scripts underlying Sections III–VII; the tools were not used to generate experimental measurements. Every line of AI-generated code was reviewed by at least one author, and the authors take responsibility for the correctness and originality of the paper, code, and results.

REFERENCES

[1] L. Patel, S. Jha, C. Guestrin, and M. Zaharia, "LOTUS: Enabling semantic queries with LLMs over tables of unstructured and structured data," arXiv preprint arXiv:2407.11418, 2024.

[2] C. Liu, M. Russo, M. Cafarella, L. Cao, P. B. Chen, Z. Chen, M. Franklin, T. Kraska, S. Madden, R. Shahout, and G. Vitagliano, "Palimpzest: Optimizing AI-powered analytics with declarative query processing," in *Conference on Innovative Data Systems Research (CIDR)*, 2025.

[3] S. Jo and I. Trummer, "ThalamusDB: Approximate query processing on multi-modal data," *Proceedings of the ACM on Management of Data*, vol. 2, no. 3, pp. 1–26, 2024.

[4] M. Urban and C. Binnig, "CAESURA: Language models as multi-modal query planners," in *Conference on Innovative Data Systems Research (CIDR)*, 2024.

[5] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.

[6] W. Chen, H. Hu, X. Chen, P. Verga, and W. W. Cohen, "MuRAG: Multimodal retrieval-augmented generator for open question answering over images and text," in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022, pp. 5558–5570.

[7] M. Yasunaga, A. Aghajanyan, W. Shi, R. James, J. Leskovec, P. Liang, M. Lewis, L. Zettlemoyer, and W.-t. Yih, "Retrieval-augmented multimodal language modeling," in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023, pp. 39 804–39 833.

[8] Amazon Web Services, "Amazon s3 storage classes," https://docs.aws.amazon.com/AmazonS3/latest/userguide/storage-class-intro.html, 2024.

[9] Google Cloud, "Cloud storage classes," https://cloud.google.com/storage/docs/storage-classes, 2024.

[10] Microsoft Azure, "Azure blob storage: Hot, cool, and archive tiers," https://learn.microsoft.com/en-us/azure/storage/blobs/access-tiers-overview, 2024.

[11] S. Lonn, P. Radeva, and M. Dimiccoli, "Smartphone picture organization: A hierarchical approach," 2019. [Online]. Available: https://arxiv.org/abs/1803.05940

[12] M. Rahnemoonfar, T. Chowdhury, A. Sarkar, D. Varshney, M. Yari, and R. R. Murphy, "FloodNet: A high resolution aerial imagery dataset for post flood scene understanding," *IEEE Access*, vol. 9, pp. 89 644–89 654, 2021.

[13] L. Chen, M. Zaharia, and J. Zou, "FrugalGPT: How to use large language models while reducing cost and improving performance," *arXiv preprint arXiv:2305.05176*, 2023.

[14] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. E. Gonzalez, H. Zhang, and I. Stoica, "Efficient memory management for large language model serving with PagedAttention," in *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, 2023, pp. 611–626.

[15] T. Xu, L. M. Botelho, and F. X. Lin, "VStore: A data store for analytics on large videos," in *Proceedings of the Fourteenth EuroSys Conference*, 2019, pp. 1–17.

[16] B. Haynes, M. Daum, D. He, A. Mazumdar, M. Balazinska, A. Cheung, and L. Ceze, "VSS: A storage system for video analytics," in *Proceedings of the 2021 International Conference on Management of Data (SIGMOD)*, 2021, pp. 685–696.

[17] M. Daum, B. Haynes, D. He, A. Mazumdar, and M. Balazinska, "TASM: A tile-based storage manager for video analytics," in *Proceedings of the IEEE 37th International Conference on Data Engineering (ICDE)*, 2021, pp. 1775–1786.

[18] D. Kang, J. Emmons, F. Abuzaid, P. Bailis, and M. Zaharia, "NoScope: Optimizing neural network queries over video at scale," *Proceedings of the VLDB Endowment*, vol. 10, no. 11, pp. 1586–1597, 2017.

[19] D. Kang, P. Bailis, and M. Zaharia, "BlazeIt: Optimizing declarative aggregation and limit queries for neural network-based video analytics," *Proceedings of the VLDB Endowment*, vol. 13, no. 4, pp. 533–546, 2019.

[20] Y. Lu, A. Chowdhery, S. Kandula, and S. Chaudhuri, "Accelerating machine learning inference with probabilistic predicates," in *Proceedings of the 2018 International Conference on Management of Data (SIGMOD)*, 2018, pp. 1493–1508.

[21] D. Kang, J. Guibas, P. Bailis, T. Hashimoto, and M. Zaharia, "Semantic indexes for machine learning-based queries over unstructured data," in *Proceedings of the 2022 International Conference on Management of Data (SIGMOD)*, 2022, pp. 1023–1036.

[22] D. Kang, A. Mathur, T. Veeramacheneni, P. Bailis, and M. Zaharia, "Jointly optimizing preprocessing and inference for DNN-based visual analytics," *Proceedings of the VLDB Endowment*, vol. 14, no. 2, pp. 87–100, 2020.

[23] L. Chen, M. Zaharia, and J. Zou, "FrugalML: How to use ML prediction APIs more accurately and cheaply," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 10 685–10 696.

[24] I. Ong, A. Almahairi, V. Wu, W.-L. Chiang, T. Wu, J. E. Gonzalez, M. W. Kadous, and I. Stoica, "RouteLLM: Learning to route LLMs with preference data," *arXiv preprint arXiv:2406.18665*, 2024.

[25] Amazon Web Services, "Amazon S3 intelligent-tiering storage class," https://aws.amazon.com/s3/storage-classes/intelligent-tiering/, 2024.

[26] V. Harinarayan, A. Rajaraman, and J. D. Ullman, "Implementing data cubes efficiently," in *Proceedings of the 1996 ACM SIGMOD International Conference on Management of Data*, 1996, pp. 205–216.

[27] S. Chaudhuri and V. R. Narasayya, "An efficient, cost-driven index selection tool for Microsoft SQL Server," in *Proceedings of the 23rd International Conference on Very Large Data Bases (VLDB)*, 1997, pp. 146–155.

[28] P. Sinha and A. A. Zoltners, "The multiple-choice knapsack problem," *Operations Research*, vol. 27, no. 3, pp. 503–515, 1979.

[29] Y. Goyal, T. Khot, D. Summers-Stay, D. Batra, and D. Parikh, "Making the V in VQA matter: Elevating the role of image understanding in visual question answering," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 6904–6913.

[30] D. A. Hudson and C. D. Manning, "GQA: A new dataset for real-world visual reasoning and compositional question answering," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 6700–6709.

[31] A. Singh, V. Natarajan, M. Shah, Y. Jiang, X. Chen, D. Batra, D. Parikh, and M. Rohrbach, "Towards VQA models that can read," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 8317–8326.

[32] P. Agrawal, S. Antoniak, E. B. Hanna, B. Bout, D. Chaplot, J. Chudnovsky, D. Costa, B. De Monicault, S. Garg, T. Gervet *et al.*, "Pixtral 12b," *arXiv preprint arXiv:2410.07073*, 2024.

[33] Meta AI, "The Llama 4 herd: The beginning of a new era of natively multimodal AI innovation," https://ai.meta.com/blog/llama-4-multimodal-intelligence/, 2025.

[34] H. Liu, C. Li, Y. Li, and Y. J. Lee, "Improved baselines with visual instruction tuning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 26 296–26 306.

[35] P. Wang, S. Bai, S. Tan, S. Wang, Z. Fan, J. Bai, K. Chen, X. Liu, J. Wang, W. Ge *et al.*, "Qwen2-VL: Enhancing vision-language model's perception of the world at any resolution," *arXiv preprint arXiv:2409.12191*, 2024.

[36] S. Bai, K. Chen, X. Liu *et al.*, "Qwen2.5-VL technical report," arXiv preprint arXiv:2502.13923, 2025.

[37] Qwen Team, "Qwen3-VL," https://github.com/QwenLM/Qwen3-VL, 2025.

[38] W. Sentosa, B. Chandrasekaran, P. B. Godfrey, and H. Hassanieh, "CellReplay: Towards accurate record-and-replay for cellular networks," in *22nd USENIX Symposium on Networked Systems Design and Implementation (NSDI 25)*. Philadelphia, PA: USENIX Association, apr 2025, pp. 1169–1186. [Online]. Available: https://www.usenix.org/conference/nsdi25/presentation/sentosa

[39] M. Cai, J. Yang, J. Gao, and Y. J. Lee, "Matryoshka multimodal models," *arXiv preprint arXiv:2405.17430*, 2024.

[40] L. Chen, H. Zhao, T. Liu, S. Bai, J. Lin, C. Zhou, and B. Chang, "An image is worth 1/2 tokens after layer 2: Plug-and-play inference acceleration for large vision-language models," *arXiv preprint arXiv:2403.06764*, 2024.

[41] B. Haynes, A. Mazumdar, A. Alaghemandi, M. Balazinska, L. Ceze, and A. Cheung, "LightDB: A DBMS for virtual reality video," *Proceedings of the VLDB Endowment*, vol. 11, no. 10, pp. 1192–1205, 2018.

[42] S. Agarwal, B. Mozafari, A. Panda, H. Milner, S. Madden, and I. Stoica, "BlinkDB: Queries with bounded errors and bounded response times on very large data," in *Proceedings of the 8th ACM European Conference on Computer Systems (EuroSys)*, 2013, pp. 29–42.

[43] D. Ding, A. Mallick, C. Wang, R. Sim, S. Mukherjee, V. Ruhle, L. V. Lakshmanan, and A. H. Awadallah, "Hybrid LLM: Cost-efficient and quality-aware query routing," *arXiv preprint arXiv:2404.14618*, 2024.

[44] T. Shnitzer, A. Ou, M. Silva, K. Soule, Y. Sun, J. Solomon, N. Thompson, and M. Yurochkin, "Large language model routing with benchmark datasets," *arXiv preprint arXiv:2309.15789*, 2023.

[45] W. Zhou, C. Xu, T. Ge, J. McAuley, K. Xu, and F. Wei, "BERT loses patience: Fast and robust inference with early exit," *Advances in Neural Information Processing Systems*, vol. 33, pp. 18 330–18 341, 2020.

[46] T. Schuster, A. Fisch, J. Gupta, M. Dehghani, D. Bahri, V. Tran, Y. Tay, and D. Metzler, "Confident adaptive language modeling," *Advances in Neural Information Processing Systems*, vol. 35, pp. 17 456–17 472, 2022.

[47] Y. Zhao and J. Callan, "Predicting query performance using smoothed relevance feedback," in *Proceedings of the 31st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*. ACM, 2008, pp. 725–726.

[48] C. Hauff, D. Hiemstra, and F. de Jong, "Improved query difficulty prediction using statistics from related queries," in *Proceedings of the 17th ACM Conference on Information and Knowledge Management (CIKM)*. ACM, 2008, pp. 1397–1398.

[49] J. Mothe and L. Tanguy, "Linguistic features to predict query difficulty," in *ACM SIGIR 2005 Workshop on Predicting Query Difficulty*, 2005.

[50] K. Kafle and C. Kanan, "An analysis of visual question answering algorithms," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 1965–1973.

[51] A. Agrawal, D. Batra, D. Parikh, and A. Kembhavi, "Don't just assume; look and answer: Overcoming priors for visual question answering," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 4971–4980.

[52] M. Bilgic and L. Getoor, "Value of information lattice: Exploiting probabilistic independence for effective feature subset acquisition," *Journal of Artificial Intelligence Research*, vol. 41, pp. 69–95, 2011.

[53] K. Kärkkäinen, M. Kachuee, O. Goldstein, and M. Sarrafzadeh, "Cost-sensitive feature-value acquisition using feature relevance," *arXiv preprint arXiv:1912.08281*, 2019.

[54] J. Kossen, C. Cangea, E. Vértes, A. Jaegle, V. Patraucean, I. Ktena, N. Tomasev, and D. Belgrave, "Active acquisition for multimodal temporal data: a challenging decision-making task," *arXiv preprint arXiv:2211.05039*, 2022.

[55] Y. Kimhi *et al.*, "CARES: Comprehensive benchmark for vision-language model resolution selection," *arXiv preprint arXiv:2510.19496*, 2025.